

# BHARATIYA ANTARIKSH HACKATHON

## HACKATHON 2026

TEAM NAME: TEAM SOLARFORGE

TEAM LEADER: [Team Leader Name]

PROBLEM STATEMENT:

Real-time Space Weather Nowcasting and Solar Flare Forecasting using uncalibrated Level-0 orbital telemetry (ISRO Aditya-L1 SoLEXS & HEL1OS instruments).



## Team Leader

Name: [Team Leader Name]

Role: Team Leader & Domain Expert

College: [College Name]

## Team Member-1

Name: [Team Member 1 Name]

Role: Lead Architect & Developer

College: [College Name]

## Team Member-2

Name: [Team Member 2 Name]

Role: Frontend UI Designer

College: [College Name]

## Team Member-3

Name: [Team Member 3 Name]

Role: Domain Researcher

College: [College Name]

## 1. How different is it from other existing ideas?

- **Direct nowcasting on uncalibrated telemetry:** Operates instantly on raw Level-0 counts from SoLEXS & HEL1OS.
- **No latency wait:** Skips traditional post-calibration waiting times (which take hours/days) to reduce response time to <1 second.
- **Kinetic precursor engineering:** Uses 1st and 2nd derivatives (flux velocity & acceleration) instead of count thresholds.

## 2. How will it solve the problem?

- **High-throughput integration:** Uses DuckDB for out-of-core merging of massive datasets (71M+ rows) in under 3 seconds.
- **LightGBM forecasting model:** Predicts solar flares (C, M, X class) over five horizons (15m, 30m, 1h, 2h, 4h).
- **Outstanding metrics:** Achieves

## 3. USP of the proposed solution

- **End-to-End automated baseline:** Bridges the gap between uncalibrated telemetry streams and immediate emergency actions.
- **Interactive live dashboard:** Integrates warning indicators, physical flux logs, and active duration clocks into a single panel.
- **Ultra-low footprint:** Ingestion and inference engines run within a tiny RAM envelope (<40MB) suitable for edge nodes.

## Data & Processing Features

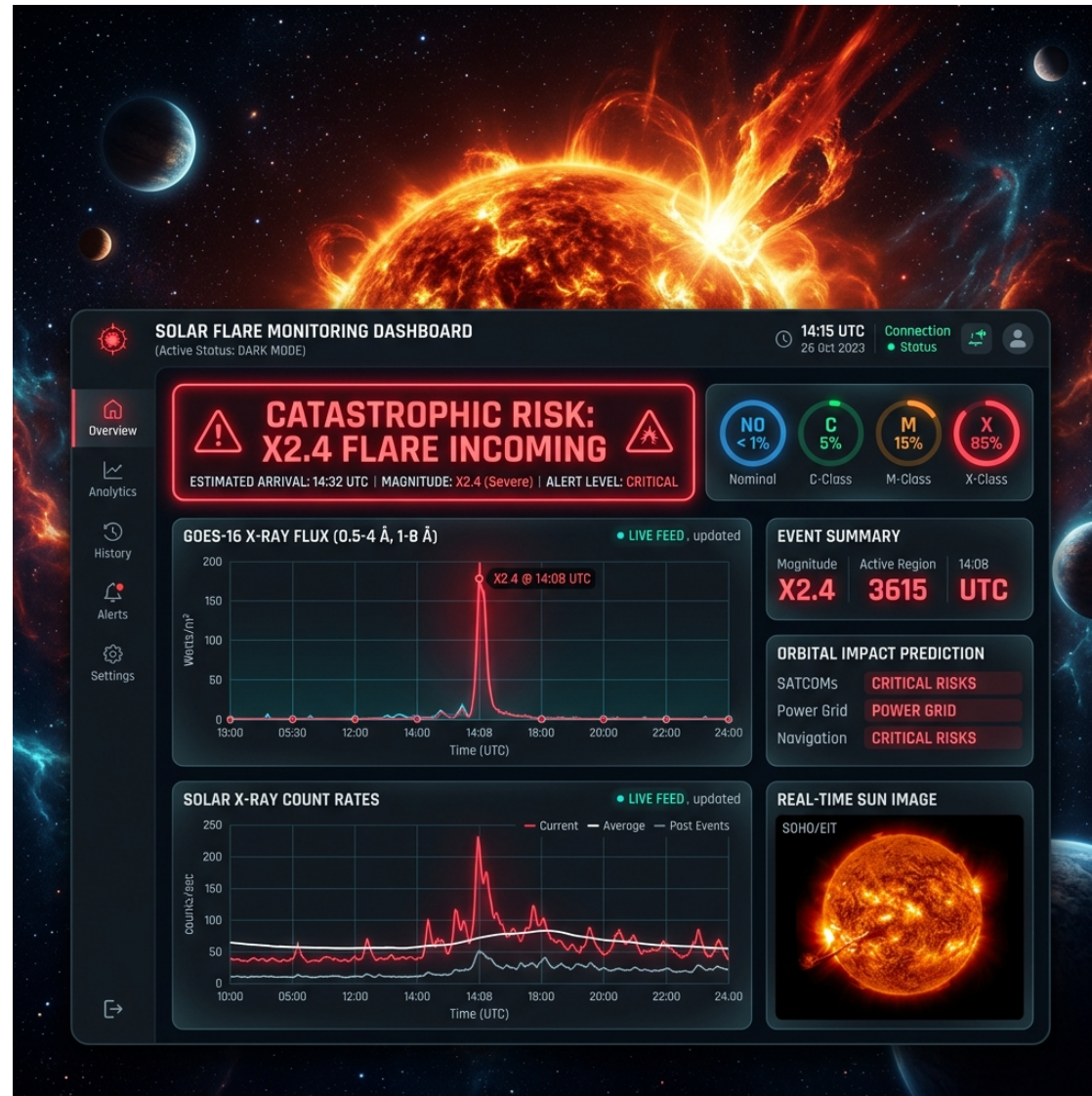
- **Out-of-Core Ingestion:** Instant querying of 71M+ records via DuckDB.
- **Telemetry Cross-Alignment:** Synchronizes thermal SoLEXS and hard disk data.
- **Float32 Compression:** Shrinks memory storage footprint by 85%.
- **Optimized Gzipped Telemetry:** Compresses database to 13.9 MB for dynamic.
- **Low-Resource Footprint:** The entire server runs on Render free instance.

## Model & UI Features

- **Multi-Horizon Lookahead:** Active alerts for +15m, +30m, +1h, +2h, and +3h.
- **High-GBM Classifier:** Predicts probabilities of imminent C, M, and X-class events.
- **Solar Flux estimation:** Calculates real-time flux ( $\text{W/m}^2$ ) in scientific notation.
- **Dynamic Warning Matrix:** Color-coordinated panels (nominal, low, moderate, high).
- **Progressive Web App:** Desktop and mobile installable with self-cleaning cache.









| Machine Learning                                                                                                                                     | Data Engineering                                                                                                                 | Backend Web Server                                                                                                       | Frontend Web App                                                                                                                     | Deployment                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• LightGBM Classifier</li><li>• Random Forest Ensembles</li><li>• Scikit-Learn &amp; SciPy Ensembles</li></ul> | <ul style="list-style-type: none"><li>• DuckDB Ingest</li><li>• Pandas DataFrames</li><li>• PyArrow Parquet Structures</li></ul> | <ul style="list-style-type: none"><li>• FastAPI Framework</li><li>• Uvicorn ASGI Engine</li><li>• Python 3.10+</li></ul> | <ul style="list-style-type: none"><li>• React SPA Framework</li><li>• Vite Builder</li><li>• Recharts Dashboard Components</li></ul> | <ul style="list-style-type: none"><li>• GitHub</li><li>• Git Version Control</li><li>• Render Web Services Hosting</li></ul> |



## Budget Breakdown

- **Software Licenses:** \$0 -- All technologies (Python, DuckDB, LightGBM, React) are open-source and free.
- **Data Ingestion APIs:** \$0 -- Free, open-access telemetry streams from ISRO and NOAA space portals.
- **Web Server Hosting:** \$7 / month -- Host backend and database on standard Render Virtual Servers.
- **Frontend Application Hosting:** \$0 -- Free static client hosting using Render or Vercel static pages.
- **Database Storage Cost:** \$0 -- Telemetry compressed to 13.9 MB, staying far below git storage limits.

**TOTAL LAUNCH BUDGET: \$7.00 / month (Fully functional, production-ready system)**

# THANK YOU

**Team SolarForge**

Live Webpage: <https://solar-flare-prediction-bscp.onrender.com>

Source Code: [https://github.com/\[GitHub Username\]/\[Repository Name\]](https://github.com/[GitHub Username]/[Repository Name])